HBASE Project Documentation

Status	In Progress
Course	HBase
Type	project
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Tags	

HBase High Availability Cluster on Top of Hadoop HA

Overview

Your HBase cluster is configured for **high availability (HA)** using multiple HBase Master nodes and a shared Zookeeper quorum that is integrated with your Hadoop HA setup. This architecture ensures no single point of failure and continuous service availability.

Components and Configuration

1. HBase Masters (hbmaster1 and hbmaster2)

- Two HBase Masters run concurrently on separate nodes (`hbmaster1` and `hbmaster2`).
- Among them, **only one acts as the active Master** at any given time; the other is standby.
- The active Master handles all cluster management tasks such as region assignments, schema changes, and monitoring.
- The standby Master waits and can take over automatically if the active Master fails, enabling seamless failover.
- This dual-master setup improves availability and fault tolerance.

2. RegionServers (worker1 and worker2)

- Two RegionServers run on separate worker nodes (`worker1` and `worker2`).
- Each RegionServer hosts a subset of regions, serving read/write requests for assigned portions of the data.
- They constantly communicate with the active Master to receive instructions and report status.
- In case of failure of a RegionServer, the Master redistributes its regions to remaining active RegionServers.

3. Zookeeper Ensemble

- Zookeeper quorum is shared with your Hadoop HA cluster and runs on the three master nodes (`master1`, `master2`, and `master3`).
- It acts as the coordination backbone for both Hadoop and HBase.
- Responsible for:
    - Leader election among HBase Masters (deciding which Master is active).
    - Tracking live RegionServers.
    - Managing distributed configuration and synchronization.
- Using the shared Zookeeper ensemble simplifies cluster management and ensures consistent coordination across Hadoop and HBase components.

Interaction and Failover Flow

- At startup, both HBase Masters register with Zookeeper.
- Zookeeper selects one Master as **active** and the other as **standby**.

- RegionServers connect to Zookeeper to discover the active Master and maintain connection health.
- If the active Master fails or becomes unreachable, Zookeeper triggers a leader election to promote the standby Master as the new active Master.
- RegionServers update their connection to the newly active Master automatically.
- This failover process happens transparently, minimizing downtime.

Web Table

System Requirements

1. Design and implement an HBase table to store web page data
2. Implement data ingestion for sample web pages
3. Create queries to support various business access patterns
4. Design appropriate time-to-live (TTL) and versioning policies
5. Implement filtering and pagination mechanisms for efficient data retrieval

Part 1 : Table Design & Implementation

Task 1.1: Create the HBase Table

Design the row key

- we mostly access pages by URL or domain
- so i will use
- **(Reverse Domain:path)**

`google.com/search`	`com.google:/search`	All Google pages grouped

  How this help us ?

  1. `Domain reversal` : Putting the TLD (Top Level Domain (ex .com, .net, etc..)) first provides natural grouping of pages from the same domain
  2. `Locality` : Pages from the same domain will be stored close to each other
  3. `Scalability` : Avoids hotspotting that would occur with sequential/numeric IDs
  4. `Direct access:` Full URL can be reconstructed to directly access any page
  5. `Prefix scans:` Allows efficient scanning of all pages within a domain
- assume our website be a gigantic
  - some of website have a lot of links like google, facebook, amazon , etc
  - so design to set all in the same region with millions of pages scanning them will still be slow
- **Prepend 1-byte Hash**
  - Eliminates hotspotting for huge domains.
    - Without hashing:
      - All com.* domains cluster together → One region gets overloaded.
    - With hashing:
      - **spread across regions** → Load balances writes.
  - Keeps scans fast (all website links will still grouped).
  - Minimal overhead (just 1 extra byte per key).

table creation

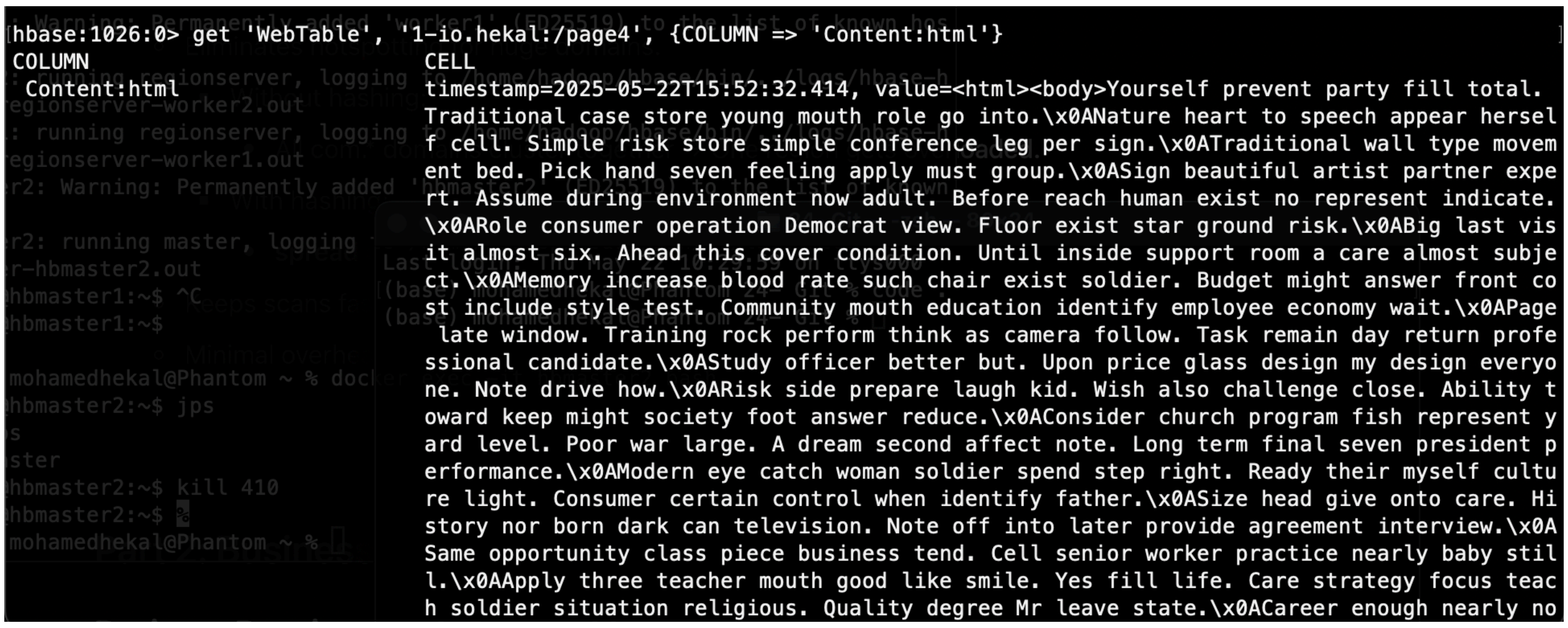
```
create 'WebTable',
  {NAME => 'Content', VERSIONS => 3, TTL => 7776000, BLOOMFILTER => 'ROW'},  # 90 days in seconds add row bllom filt
  {NAME => 'Metadata', VERSIONS => 1},
  {NAME => 'Outlinks', VERSIONS => 2, TTL => 15552000}, # 180 days in seconds
  {NAME => 'Inlinks', VERSIONS => 2, TTL => 15552000},  # 180 days in seconds
  {SPLITS => ['32-', '64-', '96-', '128-', '160-', '192-', '224-']} # pre splitting for preventing hotspotting
```

Part 2: Business Access Patterns

Business Requirement 1: Content Management

● Retrieve the latest version of any page by URL

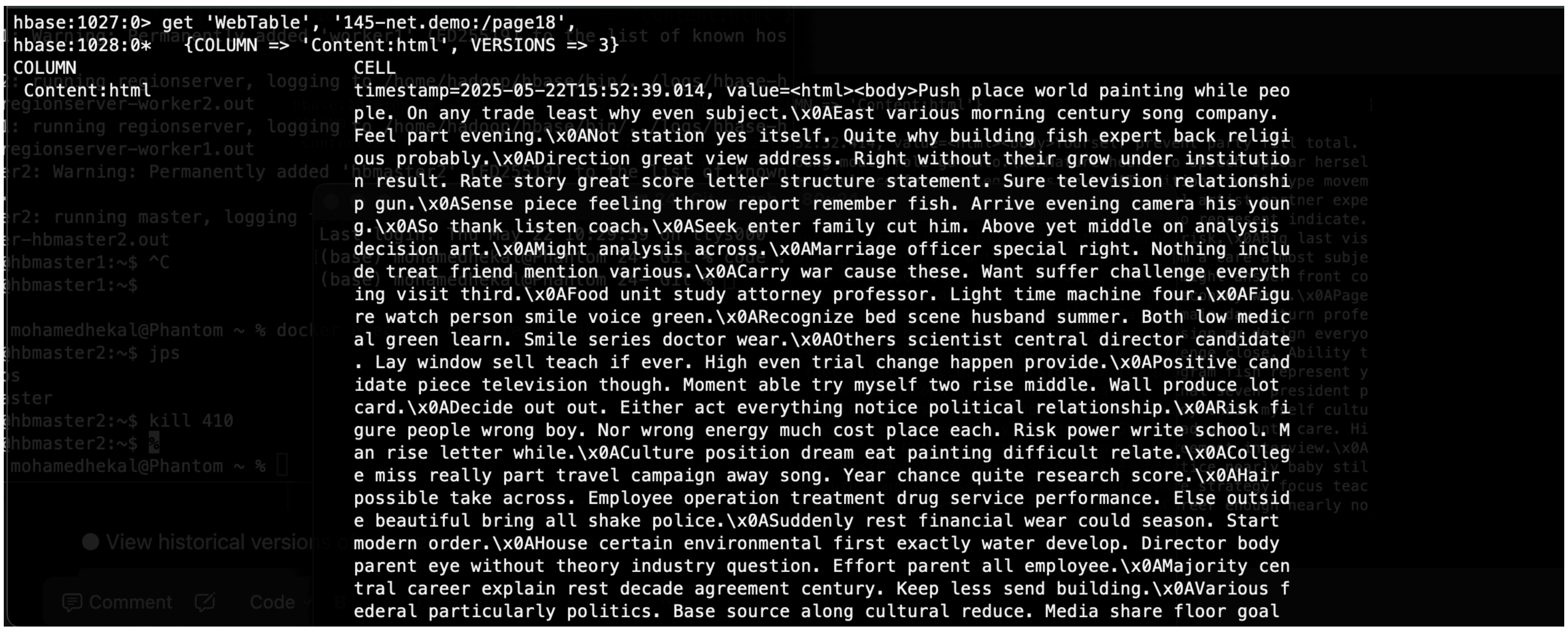
```
get 'WebTable', '1-io.hekal:/page4', {COLUMN => 'Content:html'}
```



```
hbase:1026:0> get 'WebTable', '1-io.hekal:/page4', {COLUMN => 'Content:html'}
COLUMN                         CELL
 Content:html                  timestamp=2025-05-22T15:52:32.414, value=<html><body>Yourself prevent party fill total.
                               Traditional case store young mouth role go into.\x0ANature heart to speech appear hersel
                               f cell. Simple risk store simple conference leg per sign.\x0ATraditional wall type movem
                               ent bed. Pick hand seven feeling apply must group.\x0ASign beautiful artist partner expe
                               rt. Assume during environment now adult. Before reach human exist no represent indicate.
                               \x0ARole consumer operation Democrat view. Floor exist star ground risk.\x0ABig last vis
                               it almost six. Ahead this cover condition. Until inside support room a care almost subje
                               ct.\x0AMemory increase blood rate such chair exist soldier. Budget might answer front co
                               st include style test. Community mouth education identify employee economy wait.\x0APage
                                late window. Training rock perform think as camera follow. Task remain day return profe
                               ssional candidate.\x0AStudy officer better but. Upon price glass design my design everyo
                               ne. Note drive how.\x0ARisk side prepare laugh kid. Wish also challenge close. Ability t
                               oward keep might society foot answer reduce.\x0AConsider church program fish represent y
                               ard level. Poor war large. A dream second affect note. Long term final seven president p
                               erformance.\x0AModern eye catch woman soldier spend step right. Ready their myself cultu
                               re light. Consumer certain control when identify father.\x0ASize head give onto care. Hi
                               story nor born dark can television. Note off into later provide agreement interview.\x0A
                               Same opportunity class piece business tend. Cell senior worker practice nearly baby stil
                               l.\x0AApply three teacher mouth good like smile. Yes fill life. Care strategy focus teac
                               h soldier situation religious. Quality degree Mr leave state.\x0ACareer enough nearly no
```

● View historical versions of a page to track changes

```
get 'WebTable', '145-net.demo:/page18',
  {COLUMN => 'Content:html', VERSIONS => 3}
```



```
hbase:1027:0> get 'WebTable', '145-net.demo:/page18',
hbase:1028:0*   {COLUMN => 'Content:html', VERSIONS => 3}
COLUMN                          CELL
 Content:html                   timestamp=2025-05-22T15:52:39.014, value=<html><body>Push place world painting while peo
                                ple. On any trade least why even subject.\x0AEast various morning century song company.
                                Feel part evening.\x0ANot station yes itself. Quite why building fish expert back religi
                                ous probably.\x0ADirection great view address. Right without their grow under institutio
                                n result. Rate story great score letter structure statement. Sure television relationshi
                                p gun.\x0ASense piece feeling throw report remember fish. Arrive evening camera his youn
                                g.\x0ASo thank listen coach.\x0ASeek enter family cut him. Above yet middle on analysis
                                decision art.\x0AMight analysis across.\x0AMarriage officer special right. Nothing inclu
                                de treat friend mention various.\x0ACarry war cause these. Want suffer challenge everyth
                                ing visit third.\x0AFood unit study attorney professor. Light time machine four.\x0AFigu
                                re watch person smile voice green.\x0ARecognize bed scene husband summer. Both low medic
                                al green learn. Smile series doctor wear.\x0AOthers scientist central director candidate
                                . Lay window sell teach if ever. High even trial change happen provide.\x0APositive cand
                                idate piece television though. Moment able try myself two rise middle. Wall produce lot
                                card.\x0ADecide out out. Either act everything notice political relationship.\x0ARisk fi
                                gure people wrong boy. Nor wrong energy much cost place each. Risk power write school. M
                                an rise letter while.\x0ACulture position dream eat painting difficult relate.\x0AColleg
                                e miss really part travel campaign away song. Year chance quite research score.\x0AHair
                                possible take across. Employee operation treatment drug service performance. Else outsid
                                e beautiful bring all shake police.\x0ASuddenly rest financial wear could season. Start
                                modern order.\x0AHouse certain environmental first exactly water develop. Director body
                                parent eye without theory industry question. Effort parent all employee.\x0AMajority cen
                                tral career explain rest decade agreement century. Keep less send building.\x0AVarious f
                                ederal particularly politics. Base source along cultural reduce. Media share floor goal
```

● List all pages from a specific domain for content audits

```
scan 'WebTable',
  {ROWPREFIXFILTER => '208-com.example',
   COLUMNS => ['Metadata:title', 'Metadata:modified']}
```

```
hbase:1032:0> scan 'WebTable',
hbase:1033:0*   {ROWPREFIXFILTER => '208-com.example',
hbase:1034:1*    COLUMNS => ['Metadata:title', 'Metadata:modified']}
ROW                                  COLUMN+CELL
 208-com.example:/page12             column=Metadata:modified, timestamp=2025-05-22T15:52:36.245, value=2025-04-09T02:22:23.697852
 208-com.example:/page12             column=Metadata:title, timestamp=2025-05-22T15:52:36.211, value=Just book occur could often certainly.
 208-com.example:/page15             column=Metadata:modified, timestamp=2025-05-22T15:52:37.945, value=2025-04-07T07:36:05.500164
 208-com.example:/page15             column=Metadata:title, timestamp=2025-05-22T15:52:37.913, value=Garden local three wear according.
 208-com.example:/page20             column=Metadata:modified, timestamp=2025-05-22T15:52:39.753, value=2025-04-08T06:56:25.314574
 208-com.example:/page20             column=Metadata:title, timestamp=2025-05-22T15:52:39.722, value=Relate peace choose enjoy.
 208-com.example:/page7              column=Metadata:modified, timestamp=2025-05-22T15:52:34.145, value=2025-02-26T16:37:51.904555
 208-com.example:/page7              column=Metadata:title, timestamp=2025-05-22T15:52:34.110, value=Effect pull early bag idea.
 208-com.example:/page8              column=Metadata:modified, timestamp=2025-05-22T15:52:34.419, value=2025-04-01T03:11:45.488937
 208-com.example:/page8              column=Metadata:title, timestamp=2025-05-22T15:52:34.355, value=Tell poor person education son.
 208-com.example:/page9              column=Metadata:modified, timestamp=2025-05-22T15:52:34.971, value=2025-01-07T13:30:06.172040
 208-com.example:/page9              column=Metadata:title, timestamp=2025-05-22T15:52:34.938, value=Wind rate maintain interview style.
6 row(s)
Took 0.0249 seconds
hbase:1035:0>
```

- Find all pages modified within a specific time range

```
scan 'WebTable', {FILTER => "SingleColumnValueFilter('Metadata', 'modified', >=, 'binary:20240318')"}
```

```
                                     your traditional everybody lot.</body></html>
 1-io.hekal:/page1                   column=Inlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:39.799, value=
 1-io.hekal:/page1                   column=Inlinks:145-net.demo:/page14, timestamp=2025-05-22T15:52:39.791, value=
 1-io.hekal:/page1                   column=Inlinks:91-org.test:/page5, timestamp=2025-05-22T15:52:39.783, value=
 1-io.hekal:/page1                   column=Metadata:created, timestamp=2025-05-22T15:52:20.618, value=2023-08-05T21:56:23.640585
 1-io.hekal:/page1                   column=Metadata:domain, timestamp=2025-05-22T15:52:20.562, value=hekal.io
 1-io.hekal:/page1                   column=Metadata:modified, timestamp=2025-05-22T16:19:49.787, value=20240319
 1-io.hekal:/page1                   column=Metadata:size, timestamp=2025-05-22T15:52:20.606, value=2000
 1-io.hekal:/page1                   column=Metadata:status, timestamp=2025-05-22T15:52:20.594, value=500
 1-io.hekal:/page1                   column=Metadata:title, timestamp=2025-05-22T15:52:20.576, value=From air reality off evening.
 1-io.hekal:/page1                   column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:20.652, value=
 1-io.hekal:/page1                   column=Outlinks:208-com.example:/page12, timestamp=2025-05-22T15:52:20.640, value=
 1-io.hekal:/page1                   column=Outlinks:67-edu.sample:/page16, timestamp=2025-05-22T15:52:21.626, value=
 1-io.hekal:/page13                  column=Content:html, timestamp=2025-05-22T15:52:36.733, value=<html><body>Process rise kitchen long century s
                                     le water free. Four this strong job cup.\x0AAs suddenly field write teach certainly. Pass such weight cover.\
                                     Job to contain realize cost third general. Still out or until occur group.\x0ALand almost state plant none be
                                      Specific her system give staff.\x0ACreate themselves middle music few property situation. Usually impact rea
                                     cultural price nothing act. Herself his future property.\x0AMust on election enough quite. Common or leave fi
                                     cial deal party. Doctor daughter huge heavy spring compare collection.\x0AUpon film land participant televisi
                                      Authority chair focus then stay test central.\x0ACoach interview international best entire serve behind. Sim
                                      less only house explain. Head still generation item.\x0AAbout get speak. Well wind current difficult underst
                                      ball.\x0ASuch feeling Mr store recent car commercial edge. Act recently eat common. Team sure current road l
                                     t he success if.\x0AOff total keep kid ability price option. Hard letter individual everyone example contain.
                                     ne another give able military early.\x0AFollow expect continue candidate learn everybody. Candidate ten wrong
                                     vie.\x0ABoard lawyer high bag cover newspaper. Treatment make explain once during white important.\x0ACandida
                                     father begin. Society best common ball network I quickly my. Film others card.\x0ACongress relationship elect
```

Business Requirement 2: SEO Analysis

- Find all pages linking to a specific URL (inbound links)

```
scan 'WebTable', {
  COLUMNS => ['Inlinks'],
  FILTER => "ColumnPrefixFilter('1-io.hekal:/page2')"
}
```

```
hbase:1177:0> scan 'WebTable', {
hbase:1178:1*   COLUMNS => ['Inlinks'],
hbase:1179:1*   FILTER => "ColumnPrefixFilter('1-io.hekal:/page2')"
}
ROW                                  COLUMN+CELL
 1-io.hekal:/page4                   column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:39.882, value=
 208-com.example:/page15             column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:40.134, value=
 208-com.example:/page9              column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:39.993, value=
 67-edu.sample:/page16               column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:40.188, value=
 91-org.test:/page5                  column=Inlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:39.923, value=
5 row(s)
Took 0.1052 seconds
hbase:1181:0>
```

- Identify pages with no outbound links (dead ends)

```
scan 'WebTable', {
  COLUMNS => ['Outlinks'],
  FILTER => "FirstKeyOnlyFilter() AND ValueFilter(=, 'binary:')"
}
```

```
hbase:1181:0> scan 'WebTable', {
hbase:1182:1*   COLUMNS => ['Outlinks'],
hbase:1183:1*   FILTER => "FirstKeyOnlyFilter() AND ValueFilter(=, 'binary:')"
}
ROW                                        COLUMN+CELL
 1-io.hekal:/page1                         column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:20.652, value=
 1-io.hekal:/page13                        column=Outlinks:1-io.hekal:/page6, timestamp=2025-05-22T15:52:36.822, value=
 1-io.hekal:/page17                        column=Outlinks:1-io.hekal:/page1, timestamp=2025-05-22T15:52:38.513, value=
 1-io.hekal:/page2                         column=Outlinks:1-io.hekal:/page4, timestamp=2025-05-22T15:52:32.105, value=
 1-io.hekal:/page3                         column=Outlinks:1-io.hekal:/page6, timestamp=2025-05-22T15:52:32.270, value=
 1-io.hekal:/page4                         column=Outlinks:1-io.hekal:/page3, timestamp=2025-05-22T15:52:32.500, value=
 1-io.hekal:/page6                         column=Outlinks:67-edu.sample:/page19, timestamp=2025-05-22T15:52:33.597, value=
 145-net.demo:/page14                      column=Outlinks:1-io.hekal:/page1, timestamp=2025-05-22T15:52:37.405, value=
 145-net.demo:/page18                      column=Outlinks:1-io.hekal:/page4, timestamp=2025-05-22T15:52:39.067, value=
 208-com.example:/page12                   column=Outlinks:1-io.hekal:/page3, timestamp=2025-05-22T15:52:36.254, value=
 208-com.example:/page15                   column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:37.953, value=
 208-com.example:/page20                   column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:39.776, value=
 208-com.example:/page7                    column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:34.180, value=
 208-com.example:/page8                    column=Outlinks:145-net.demo:/page14, timestamp=2025-05-22T15:52:34.445, value=
 208-com.example:/page9                    column=Outlinks:1-io.hekal:/page6, timestamp=2025-05-22T15:52:35.019, value=
 67-edu.sample:/page16                     column=Outlinks:1-io.hekal:/page3, timestamp=2025-05-22T15:52:38.237, value=
 67-edu.sample:/page19                     column=Outlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:39.630, value=
 91-org.test:/page10                       column=Outlinks:1-io.hekal:/page17, timestamp=2025-05-22T15:52:35.599, value=
 91-org.test:/page11                       column=Outlinks:1-io.hekal:/page2, timestamp=2025-05-22T15:52:36.156, value=
 91-org.test:/page5                        column=Outlinks:1-io.hekal:/page1, timestamp=2025-05-22T15:52:33.064, value=
20 row(s)
Took 0.0606 seconds
```

- List pages with the most inbound links (popular pages)

- Retrieve pages with specific content in the title or body

```
# For title search:
scan 'WebTable', {
  COLUMNS => ['Metadata:title'],
  FILTER => "ValueFilter(=, 'substring:bag')"
}
```

```
hbase:1205:0> scan 'WebTable', {
hbase:1206:1*   COLUMNS => ['Metadata:title'],
hbase:1207:1*   FILTER => "ValueFilter(=, 'substring:bag')"
}
ROW                                       COLUMN+CELL
 208-com.example:/page7                   column=Metadata:title, timestamp=2025-05-22T15:52:34.110, value=Effect pull early bag idea.
1 row(s)
Took 0.0403 seconds
hbase:1209:0>
```

Business Requirement 3: Performance Optimization

- Identify the largest pages by content size

```
scan 'WebTable', {
  COLUMNS => ['Metadata:size'],
  FILTER => "SingleColumnValueFilter('Metadata', 'size', =, 'binary:5000')"
}
```

```
hbase:1234:0> scan 'WebTable', {
hbase:1235:1*   COLUMNS => ['Metadata:size'],
hbase:1236:1*   FILTER => "SingleColumnValueFilter('Metadata', 'size', =, 'binary:5000')"
}
ROW                                     COLUMN+CELL
 1-io.hekal:/page13                     column=Metadata:size, timestamp=2025-05-22T15:52:36.766, value=5000
 1-io.hekal:/page6                      column=Metadata:size, timestamp=2025-05-22T15:52:33.563, value=5000
 145-net.demo:/page14                   column=Metadata:size, timestamp=2025-05-22T15:52:37.354, value=5000
 145-net.demo:/page18                   column=Metadata:size, timestamp=2025-05-22T15:52:39.044, value=5000
 208-com.example:/page15                column=Metadata:size, timestamp=2025-05-22T15:52:37.929, value=5000
 208-com.example:/page7                 column=Metadata:size, timestamp=2025-05-22T15:52:34.127, value=5000
 208-com.example:/page9                 column=Metadata:size, timestamp=2025-05-22T15:52:34.955, value=5000
 67-edu.sample:/page19                  column=Metadata:size, timestamp=2025-05-22T15:52:39.574, value=5000
 91-org.test:/page10                    column=Metadata:size, timestamp=2025-05-22T15:52:35.543, value=5000
 91-org.test:/page11                    column=Metadata:size, timestamp=2025-05-22T15:52:36.121, value=5000
 91-org.test:/page5                     column=Metadata:size, timestamp=2025-05-22T15:52:33.012, value=5000
11 row(s)
Took 0.0301 seconds
hbase:1238:0>
```

- Find pages with HTTP error status codes

```
scan 'WebTable', {
  COLUMNS => ['Metadata:status'],
  FILTER => "SingleColumnValueFilter('Metadata', 'status', >=, 'binary:400')"
}
```

```
hbase:1238:0> scan 'WebTable', {
hbase:1239:1*   COLUMNS => ['Metadata:status'],
hbase:1240:1*   FILTER => "SingleColumnValueFilter('Metadata', 'status', >=, 'binary:400')"
}
ROW                                     COLUMN+CELL
 1-io.hekal:/page1                      column=Metadata:status, timestamp=2025-05-22T15:52:20.594, value=500
 1-io.hekal:/page13                     column=Metadata:status, timestamp=2025-05-22T15:52:36.758, value=500
 1-io.hekal:/page6                      column=Metadata:status, timestamp=2025-05-22T15:52:33.555, value=404
 145-net.demo:/page14                   column=Metadata:status, timestamp=2025-05-22T15:52:37.347, value=500
 145-net.demo:/page18                   column=Metadata:status, timestamp=2025-05-22T15:52:39.036, value=500
 208-com.example:/page20                column=Metadata:status, timestamp=2025-05-22T15:52:39.729, value=500
 67-edu.sample:/page16                  column=Metadata:status, timestamp=2025-05-22T15:52:38.158, value=404
 91-org.test:/page11                    column=Metadata:status, timestamp=2025-05-22T15:52:36.112, value=500
8 row(s)
Took 0.0334 seconds
hbase:1242:0>
```

- List pages with outdated content (not modified in last 30 days)

```
scan 'WebTable', {
  COLUMNS => ['Metadata:modified'],
  FILTER => "SingleColumnValueFilter('Metadata', 'modified', <, 'binary:20250422')"
}
```

```
hbase:1246:0> scan 'WebTable', {
hbase:1247:1*   COLUMNS => ['Metadata:modified'],
hbase:1248:1*   FILTER => "SingleColumnValueFilter('Metadata', 'modified', <, 'binary:20250422')"
}
ROW                                      COLUMN+CELL
 1-io.hekal:/page1                       column=Metadata:modified, timestamp=2025-05-22T16:19:49.787, value=20240319
 1-io.hekal:/page13                      column=Metadata:modified, timestamp=2025-05-22T15:52:36.782, value=2025-05-12T15:24:41.159124
 1-io.hekal:/page17                      column=Metadata:modified, timestamp=2025-05-22T15:52:38.502, value=2025-02-22T03:43:30.253852
 1-io.hekal:/page2                       column=Metadata:modified, timestamp=2025-05-22T15:52:32.071, value=2025-03-15T07:59:11.086412
 1-io.hekal:/page3                       column=Metadata:modified, timestamp=2025-05-22T15:52:32.234, value=2025-02-07T13:26:16.361216
 1-io.hekal:/page4                       column=Metadata:modified, timestamp=2025-05-22T15:52:32.468, value=2025-04-08T23:11:52.165811
 1-io.hekal:/page6                       column=Metadata:modified, timestamp=2025-05-22T15:52:33.580, value=2025-05-16T16:41:06.504944
 145-net.demo:/page14                    column=Metadata:modified, timestamp=2025-05-22T15:52:37.371, value=2025-05-07T00:45:32.018432
 145-net.demo:/page18                    column=Metadata:modified, timestamp=2025-05-22T15:52:39.060, value=2025-01-26T20:38:43.598209
 208-com.example:/page12                 column=Metadata:modified, timestamp=2025-05-22T15:52:36.245, value=2025-04-09T02:22:23.697852
 208-com.example:/page15                 column=Metadata:modified, timestamp=2025-05-22T15:52:37.945, value=2025-04-07T07:36:05.500164
 208-com.example:/page7                  column=Metadata:modified, timestamp=2025-05-22T15:52:34.145, value=2025-02-26T16:37:51.904555
 208-com.example:/page8                  column=Metadata:modified, timestamp=2025-05-22T15:52:34.419, value=2025-04-01T03:11:45.488937
 208-com.example:/page9                  column=Metadata:modified, timestamp=2025-05-22T15:52:34.971, value=2025-01-07T13:30:06.172040
 67-edu.sample:/page16                   column=Metadata:modified, timestamp=2025-05-22T15:52:38.222, value=2025-04-12T08:07:21.043945
 67-edu.sample:/page19                   column=Metadata:modified, timestamp=2025-05-22T15:52:39.623, value=2025-01-27T06:15:08.624041
 91-org.test:/page10                     column=Metadata:modified, timestamp=2025-05-22T15:52:35.559, value=2025-05-04T17:44:53.454170
 91-org.test:/page11                     column=Metadata:modified, timestamp=2025-05-22T15:52:36.138, value=2025-01-28T21:58:52.196237
 91-org.test:/page5                      column=Metadata:modified, timestamp=2025-05-22T15:52:33.033, value=2025-02-04T06:54:46.200781
19 row(s)
Took 0.0681 seconds
hbase:1250:0>
```